

Auto-Ticket Classification using Flow Designer

Project Description:

Auto-Ticket Classification using Flow Designer is a ServiceNow-based automation project developed for a school IT helpdesk. The system automates the classification of newly created IT incident tickets by identifying relevant keywords in the ticket description.

The flow automatically assigns the appropriate Category and Subcategory, maintains the configured classification choices, stores the ticket information in the Incident record, and sends an automated email notification to the caller. The solution is designed to reduce manual classification work and improve ticket routing efficiency.

Scenarios

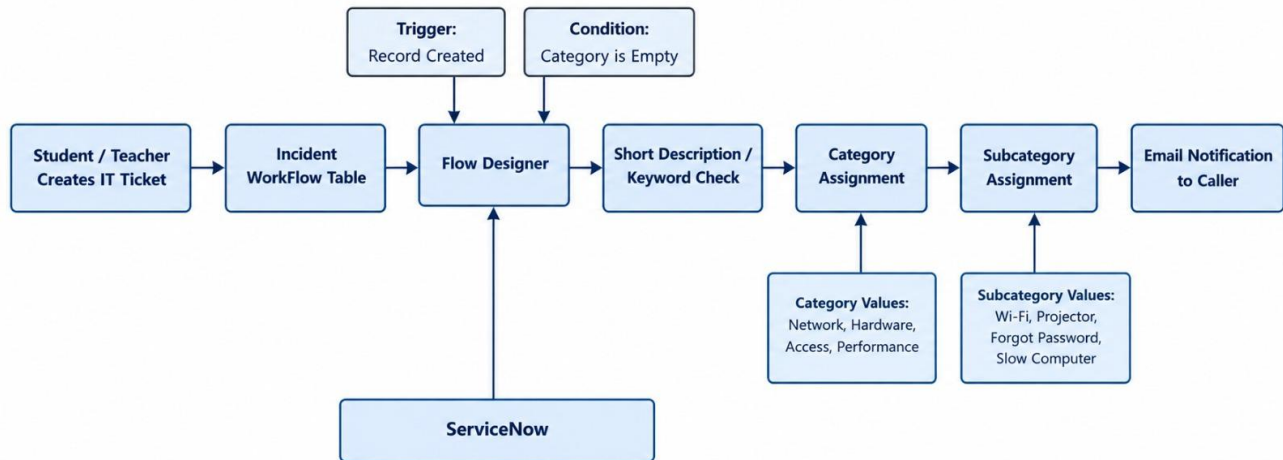
Scenario 1: A user creates an incident with the description “WiFi not working in library”. The system identifies the Wi-Fi issue and assigns Category as Network and Subcategory as Wi-Fi. An email notification is sent to the caller.

Scenario 2: A user creates an incident with the description “Projector not turning on”. The system identifies the projector issue and assigns Category as Hardware and Subcategory as Projector. An email notification is sent to the caller.

Scenario 3: The classification configuration can also represent other defined school IT issues such as Forgot Password and Slow Computer using the configured Category and Subcategory mappings.

Technical Architecture:

Auto-Ticket Classification using Flow Designer – Project Flow



Description: The Incident WorkFlow table stores ticket details such as short description, caller, Category, Subcategory, and other required information. Flow Designer acts as the automation layer. When a new record is created and Category is empty, the flow checks the ticket description, identifies the keyword, assigns Category and Subcategory, and sends the configured email notification.

Flow: New Incident → Flow Trigger → Check Description → Identify Keyword → Assign Category → Assign Subcategory → Send Email Notification

Pre-requisites:

- ServiceNow instance access
- Incident management access
- Flow Designer access
- Configured Category and Subcategory choices
- Email notification configuration
- Project Update Set

Project Workflow:

Milestone 1 – Requirement Analysis & Planning

- **Activity 1.1 – Business Use Case**
- **Activity 1.2 – Business Requirements**
- **Activity 1.3 – Creation of New Update Set**

Milestone 2 – Backend Development & Configuration

- **Activity 2.1 – Creation of Custom Table**
- **Activity 2.2 – Field Creation and Data Type Configuration**
- **Activity 2.3 – Category and Subcategory Dependency**

Milestone 3 – Automation using Flow Designer & Email Notification

- **Activity 3.1 – Create Flow**
- **Activity 3.2 – Trigger Configuration**
- **Activity 3.3 – Wi-Fi Issue Classification**
- **Activity 3.4 – Projector Issue Classification**
- **Activity 3.5 – Password Issue Classification**
- **Activity 3.6 – Slow Computer Classification**
- **Activity 3.7 – Email Action Setup**

Milestone 4 – Testing, Validation & Security

- **Activity 4.1 – Testing and Validation**
- **Activity 4.2 – Email Notification Verification**
- **Activity 4.3 – Test Scenario 2**
- **Activity 4.4 – Data Validation Checks**

Milestone 5 – Deployment & Final Documentation

- **Activity 5.1 – Make Update Set Complete**
- **Activity 5.2 – Export Update Set as XML**

Milestone 1 – Requirement Analysis & Planning

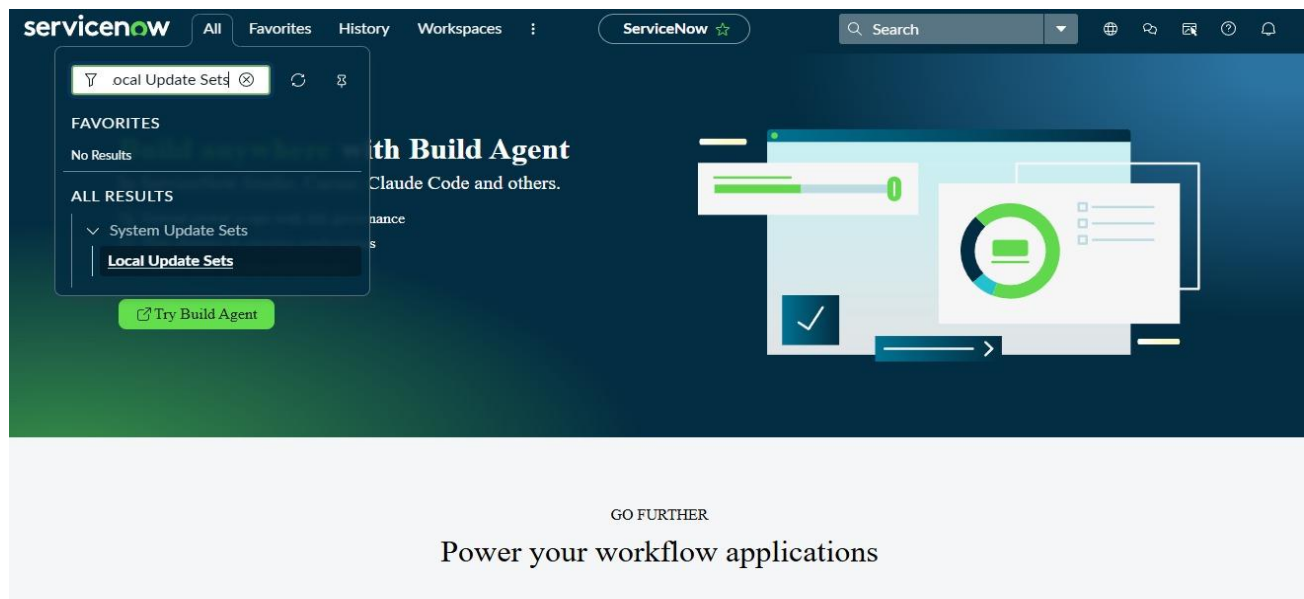
Activity 1.1 – Business Use Case

The school IT helpdesk receives multiple support requests daily from students and teachers. Common requests include network connectivity issues, hardware failures, account access problems, and system performance issues.

Currently, IT staff manually review each ticket to:

- Identify the issue type.
- Select the correct Category.
- Select the correct Subcategory.
- Inform the caller about ticket creation.

This manual process requires additional effort and can become difficult to manage when the number of support requests increases.



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Activity 1.2 – Business Requirements

The system must:

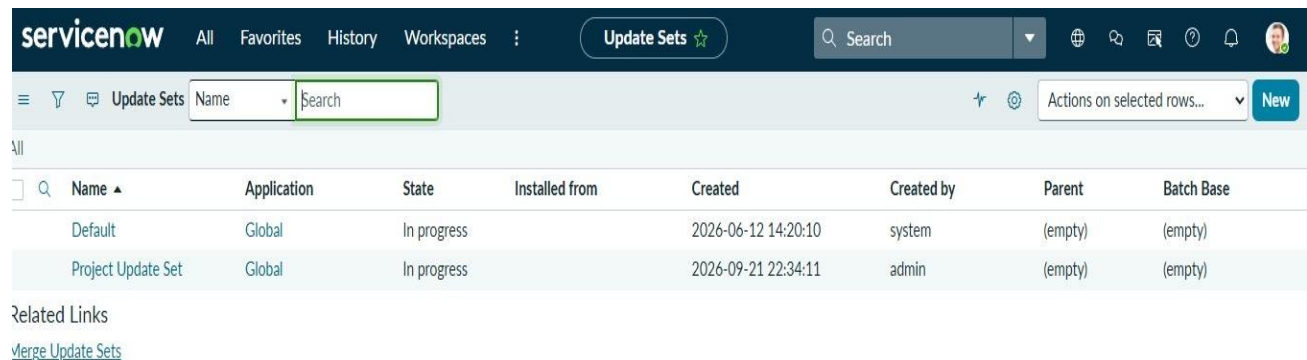
1. Automatically classify IT tickets based on the issue description.
2. Assign both Category and Subcategory without manual intervention.
3. Support dependent choice logic between Category and Subcategory.
4. Send an automated email notification to the caller upon ticket creation.
5. Store ticket information in a structured and standardized format.
6. Ensure easy maintenance and future scalability.

Activity 1.3 – Creation of New Update Set

The project configuration is maintained using a dedicated Update Set.

Navigation Steps:

- Go to **All** → **Search for Update Set**.
- Under **System Update Sets**, select **Local Update Sets**.
- Click the **New** button.
- Enter the Update Set Name as **Project Update Set**.
- Set **State**: In Progress.
- Set **Application**: Global.
- Save the form.
- Under Related Links, click **Make This My Current Set**.
- The created Update Set will then appear as the current Update Set.



The screenshot shows the 'servicenow' interface for 'Update Sets'. The top navigation bar includes 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The 'Update Sets' section is active, showing a table with the following data:

Name	Application	State	Installed from	Created	Created by	Parent	Batch Base
Default	Global	In progress		2026-06-12 14:20:10	system	(empty)	(empty)
Project Update Set	Global	In progress		2026-09-21 22:34:11	admin	(empty)	(empty)

Below the table, there is a 'Related Links' section with a link to [Merge Update Sets](#).

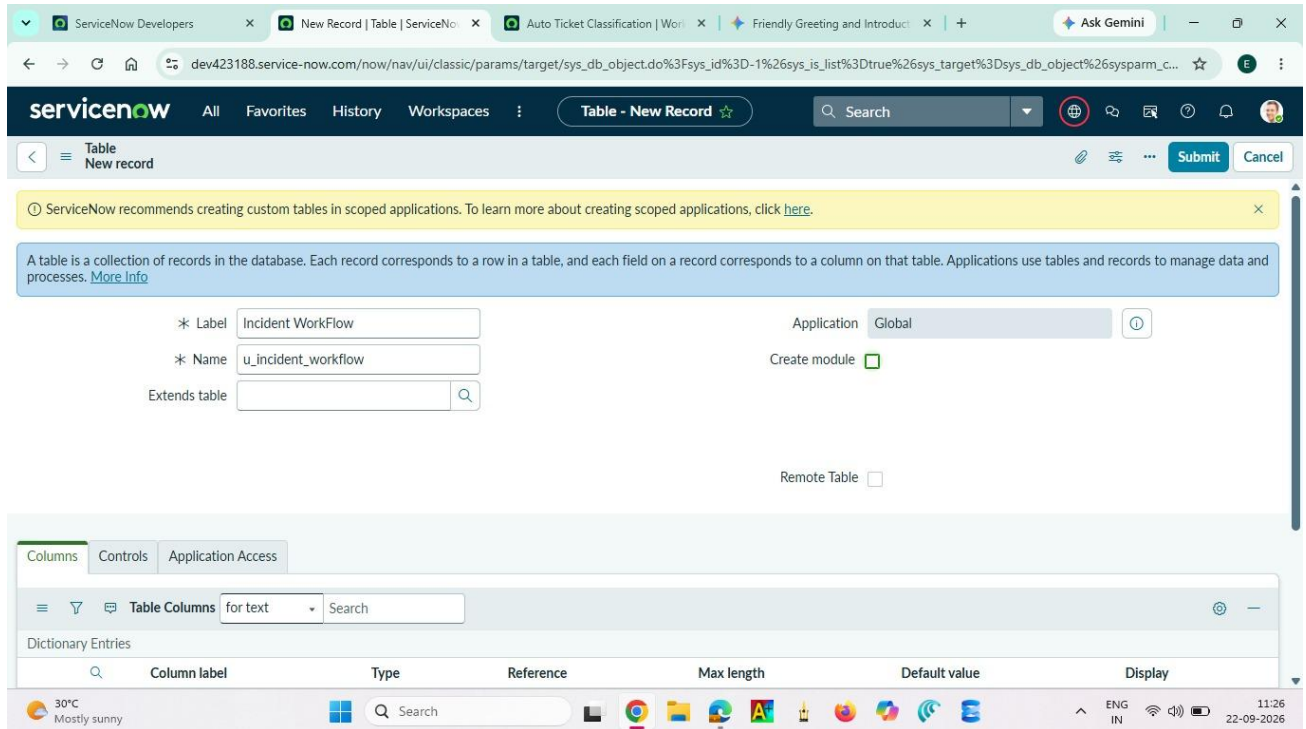
Milestone 2 – Backend Development & Configuration

Activity 2.1 – Creation of Custom Table to Store Ticket Records

A custom table named **Incident WorkFlow** is created to store the IT ticket records.

Steps to Create Custom Table:

- Open **All** → **Search for Tables**.
- Under **System Definition**, click **Tables**.
- Click the **New** button.
- Set the Label as **Incident WorkFlow**.
- Uncheck **Create module**.
- Save the form.
- Open the **Controls** section.
- Enable **Auto Number**.
- Save the form again.



The screenshot shows the ServiceNow 'Table - New Record' form. The browser address bar indicates the URL: dev423188.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sysparm_c... The form header includes 'servicenow', navigation links (All, Favorites, History, Workspaces), and a search bar. A yellow banner at the top states: 'ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).' Below this, a blue box explains: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'.

The form fields are as follows:

- * Label:** Incident WorkFlow
- * Name:** u_incident_workflow
- Extends table:** (empty field with a search icon)
- Application:** Global
- Create module:** ☐
- Remote Table:** ☐

At the bottom, there are tabs for 'Columns', 'Controls', and 'Application Access'. The 'Columns' tab is active, showing a 'Table Columns' section with a search bar and a 'Dictionary Entries' table. The table has the following columns: Column label, Type, Reference, Max length, Default value, and Display.

Activity 2.2 – Field Creation and Data Type Configuration

The required fields are created in the Incident WorkFlow table to ensure structured and consistent ticket information.

Field Label	Type	Reference / Choice
Number	Auto Number	—
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer
Short Description	String	—
Description	String	—
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

The Choice fields include **Category**, **Subcategory**, and **State**. The required choices are added through the Form Designer.



The screenshot shows the 'Form Design' interface for a table named 'Incident Workflow [u_incident_workflow]'. The interface includes a top bar with 'Incident Workflow [u_incident_workflow]', 'Default view', and 'Form Design' tabs, along with 'Undo' and 'Save' buttons. On the left, there is a 'Fields' panel with a 'reference' filter and a list of fields: 'Number', 'Caller', 'Category', 'Subcategory', 'Short Description', 'Description', 'State', 'Assigned Group', and 'Assigned to'. The main area displays the form layout for these fields, each with a configuration icon (gear) and a delete icon (X). The 'Incident Workflow [u_incident_workflow]' title bar also shows '1 Column' and a plus icon.

Activity 2.3 – Implementing Dependency Between Category and Subcategory

A dependency is implemented between Category and Subcategory so that the available Subcategory options change according to the selected Category.

Navigation Steps:

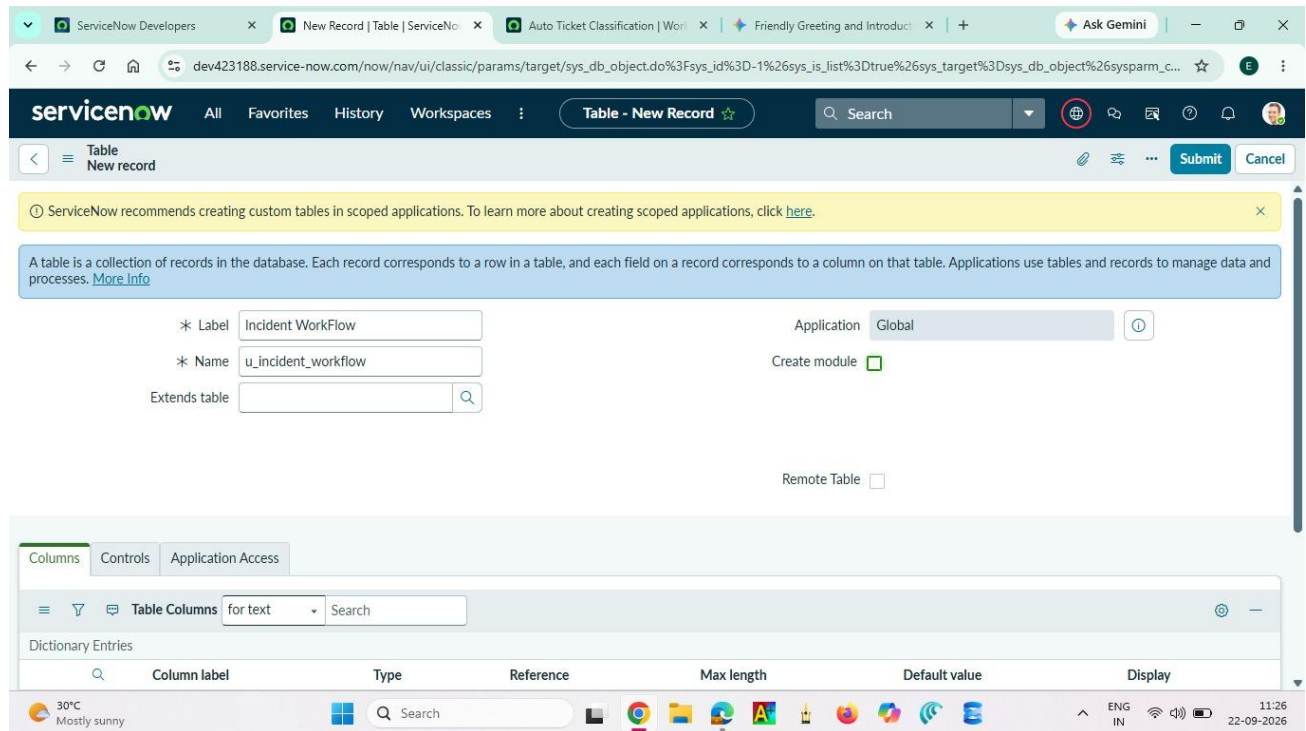
- Open the created table form.
- Right-click the **Subcategory** field.
- Select **Configure Dictionary**.
- Open **Advanced View**.
- Locate the **Dependent Field** section.
- Enable **Use dependent field**.
- Select **Category** in the **Dependent on** field.
- Save the form.

Dependent Choice Mapping:

Subcategory Dependent Category

Wi-Fi	Network
Projector	Hardware
Forgot Password	Access
Slow Computer	Performance

After configuring the dependency, the Subcategory choices are displayed according to the selected Category.



The screenshot shows the ServiceNow 'Table - New Record' form. The browser address bar indicates the URL: `dev423188.service-now.com/now/nav/ui/classic/params/target/sys_db_object.do%3Fsys_id%3D-1%26sys_is_list%3Dtrue%26sys_target%3Dsys_db_object%26sysparm_c...`. The form includes a navigation bar with 'Table - New Record' and a search bar. A yellow banner at the top states: 'ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).' Below this, a blue box explains: 'A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)'.

The form fields are as follows:

- * Label: Incident WorkFlow
- * Name: u_incident_workflow
- Extends table: (empty field with a search icon)
- Application: Global (dropdown menu)
- Create module: ☐
- Remote Table: ☐

At the bottom, there is a 'Columns' tab with a search bar and a table titled 'Dictionary Entries'.

Column label	Type	Reference	Max length	Default value	Display
Dictionary Entries					

The Windows taskbar at the bottom shows the date and time as 11:26 on 22-09-2026, along with system icons for network, volume, and battery.

Milestone 3 – Automation using Flow Designer & Email Notification

Activity 3.1 – Auto Classification using Flow Designer

The automation is implemented using ServiceNow Flow Designer.

When a student creates a new IT support ticket, the user enters the Caller and Short

Description. The user does not need to manually select Category or Subcategory.

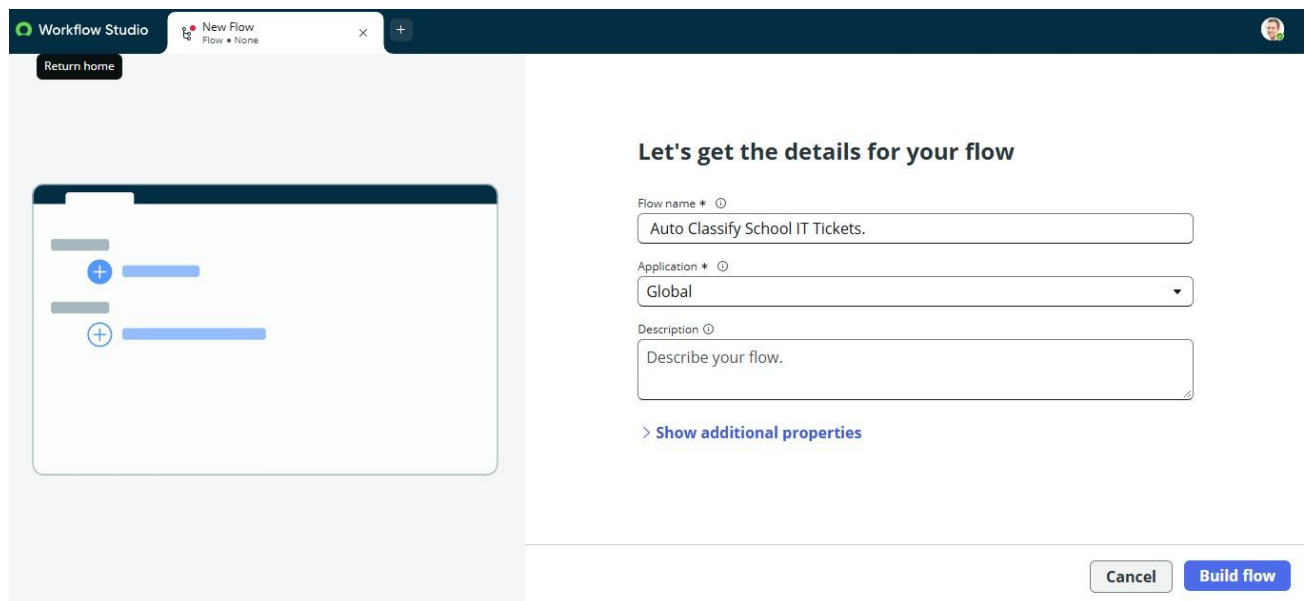
The Flow Designer evaluates the Short Description and identifies predefined keywords such as:

- WiFi
- Projector
- Password
- Login
- Slow
- Hanging

The flow then automatically assigns the appropriate Category and Subcategory.

Navigation Steps:

- Go to **All** → **Search Flow Designer**.
- Under **Process Automation**, open **Flow Designer**.
- Click **New**.
- Select **Flow**.
- Enter Flow Name: **Auto Classify School IT Tickets**.
- Set Application as **Global**.
- Click **Build Flow**.



Activity 3.2 – Trigger Configuration

The Flow is configured to execute when a new ticket record is created.

Trigger Configuration:

- **Trigger:** Record Created
- **Table:** Incident WorkFlow
- **Condition:** Category is Empty
- Click **Done**.

- Under Actions, select **Flow Logic**.

This ensures that the flow processes newly created tickets that have not yet been classified.

Activity 3.3 – Wi-Fi Issue Classification

An **If condition** is created for Wi-Fi/network-related tickets.

Condition:

- Select **If** condition.
- Condition label: **SD is WiFi or Network**.
- Select **Short Description** using the data pill picker.
- Operator: **Contains**.
- Value: **Wi-Fi**.
- Another condition can check whether Short Description contains **Network**.
- Click **Done**.

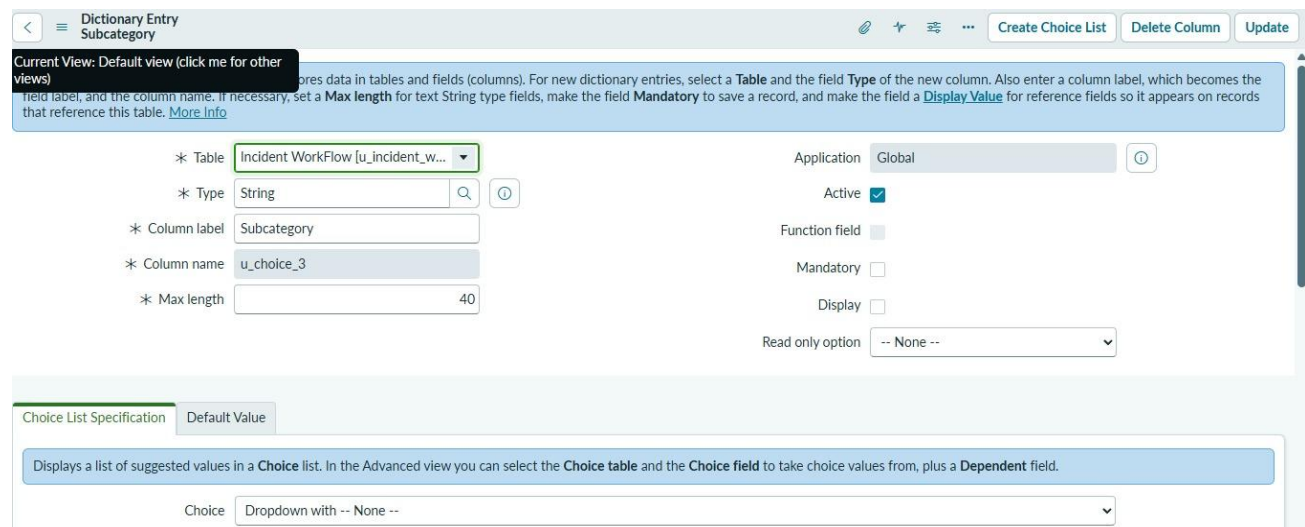
Action:

Select **Update Record**.

Set:

- **Category = Network**
- **Subcategory = Wi-Fi**

Then click **Done**.



The screenshot shows the 'Dictionary Entry Subcategory' form. At the top, there are tabs for 'Choice List Specification' and 'Default Value'. The 'Choice List Specification' tab is active, showing a text area with instructions: 'Displays a list of suggested values in a Choice list. In the Advanced view you can select the Choice table and the Choice field to take choice values from, plus a Dependent field.' Below this is a 'Choice' dropdown menu set to 'Dropdown with -- None --'. The form fields are organized into two columns. The left column contains: '* Table' (Incident Workflow [u_incident_w...]), '* Type' (String), '* Column label' (Subcategory), '* Column name' (u_choice_3), and '* Max length' (40). The right column contains: 'Application' (Global), 'Active' (checked), 'Function field' (unchecked), 'Mandatory' (unchecked), 'Display' (unchecked), and 'Read only option' (None). At the top right of the form, there are buttons for 'Create Choice List', 'Delete Column', and 'Update'.

Activity 3.4 – Projector Issue Classification

An **Else If** condition is created for projector-related issues.

Condition:

- Select **Else If**.
- Condition label: **SD is Projector or Hardware**.
- Check whether Short Description contains **Projector**.
- Click **Done**.

Action:

Select **Update Record**.

Set:

- **Category = Hardware**
- **Subcategory = Projector**

Then click **Done**.

Activity 3.5 – Password Issue Classification

An **Else If** condition is created for password-related issues.

Condition:

- Select **Else If**.
- Condition label: **SD is Forgot Password**.
- Check whether Short Description contains **Forgot Password**.
- Click **Done**.

Action:

Select **Update Record**.

Set:

- **Category = Access**
- **Subcategory = Forgot Password**

Then click **Done**.

Activity 3.6 – Slow Computer Classification

An **Else If** condition is created for performance-related issues.

Condition:

- Select **Else If**.
- Condition label: **SD is Performance**.
- Check whether Short Description contains **Slow Computer**.
- Click **Done**.

Action:

Select **Update Record**.

Set:

- **Category = Performance**
- **Subcategory = Slow Computer**

Then click **Done**.

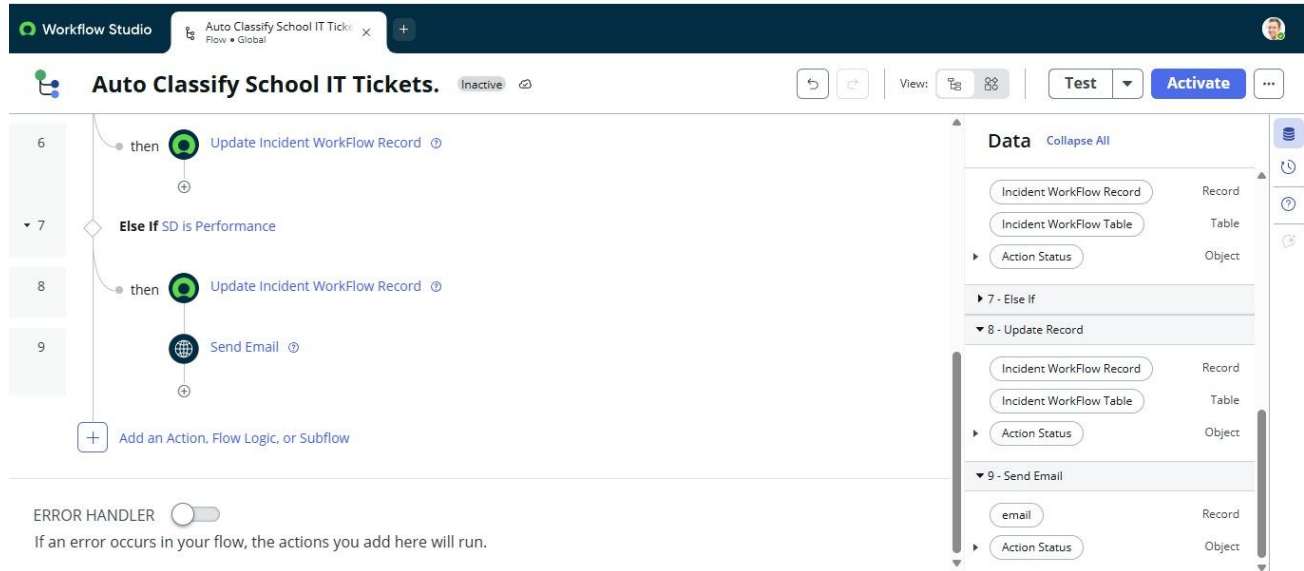
Activity 3.7 – Email Action Setup

After the ticket has been classified, an email notification is sent to the caller.

Email Action Configuration:

- Select **Send Email** action.
- **To:** Caller → Email
- **Subject:** Your Request for the issue has been submitted.
- **Body:** Ticket confirmation message.
- Click **Done**.

After completing the Flow configuration, click the **Activate** button from the top-right corner.



Milestone 4 – Testing, Validation & Security

Activity 4.1 – Testing and Validation

The testing phase verifies that the developed automation works according to the defined requirements.

The validation checks include:

- Mandatory fields are captured correctly.
- Auto-number is generated without duplication.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly to the appropriate user and group tables.
- Automatic classification works according to the configured conditions.
- Email notification is sent to the caller.

Test Scenario 1 – Wi-Fi Issue

Test Steps:

- Enter a Caller Name.
- Enter the Short Description:

“WiFi not working in library”

- Save the form.
- Reload the form.

Expected / Actual Result:

- **Category → Network**
- **Subcategory → Wi-Fi**
- **Email → Sent to Caller Email ID**

The ticket is automatically classified without manually selecting Category and Subcategory.

Activity 4.2 – Check Email Notification

Navigation Steps:

- Open **All → Search Emails**.
- Under **System Logs**, select **Emails**.
- Search using the configured email subject.
- Open the email record.
- Under Related Links, select **Preview Mail**.

This verifies that the email notification has been generated and delivered as part of the automation.

Activity 4.3 – Test Scenario 2 – Projector Issue

Test Steps:

- Enter a Caller Name.
- Enter the Short Description:

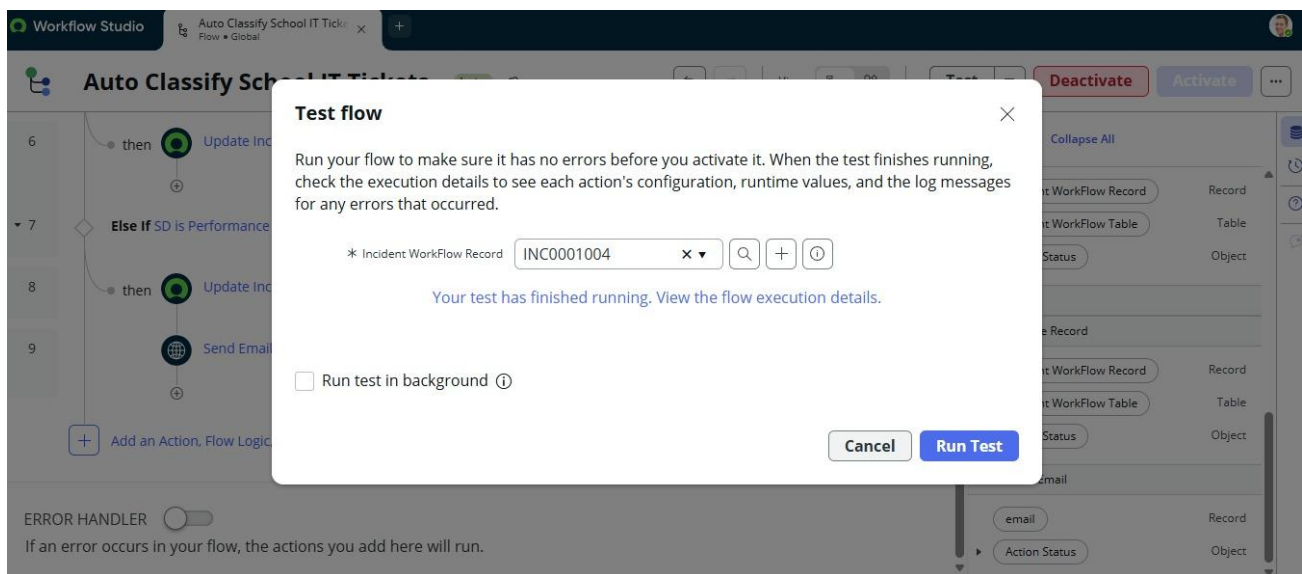
“Projector not turning on.”

- Save the form.
- Reload the form.

Expected / Actual Result:

- **Category → Hardware**
- **Subcategory → Projector**
- **Email → Sent to Caller Email ID**

The Flow Designer correctly identifies the projector-related issue and updates the ticket automatically.



Milestone 5 – Deployment & Final Documentation

Activity 5.1 – Make Update Set to Complete State

After completing the development and testing, the Update Set is moved from In Progress to Complete.

Steps:

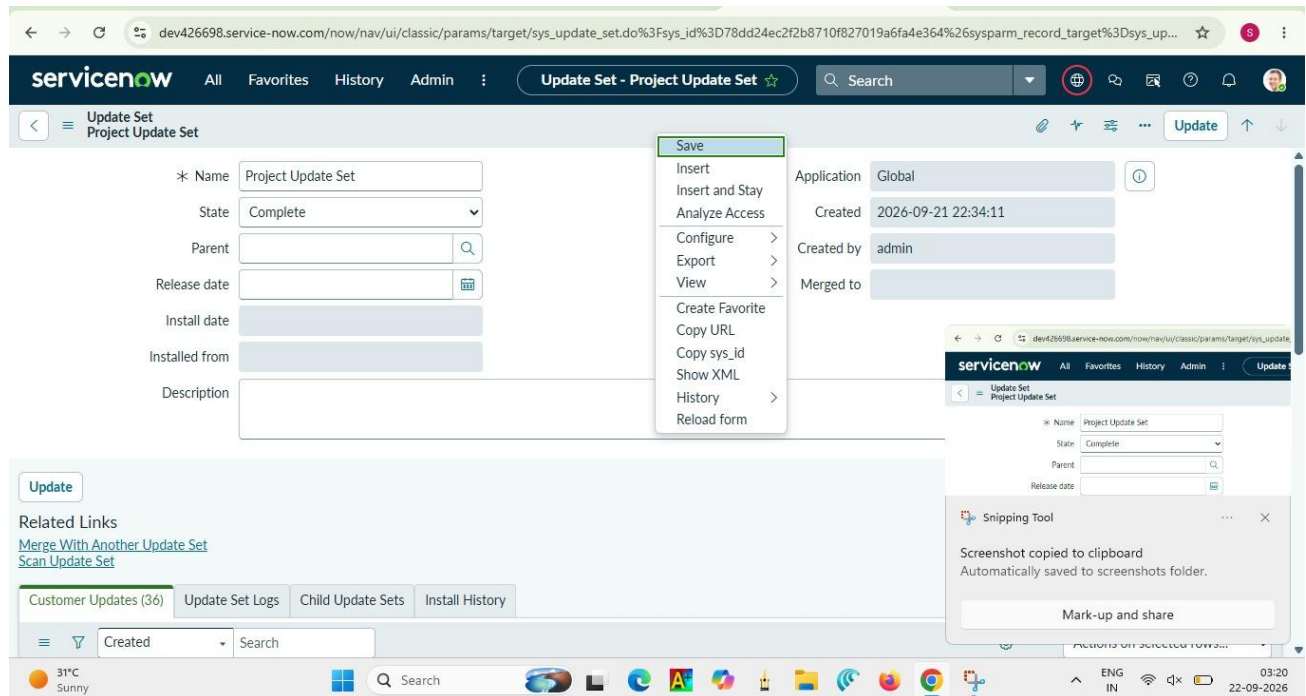
- Go to **All → Search for Local Update Sets**.
- Under **System Update Sets**, select **Local Update Sets**.
- Open **Project Update Set**.
- Change the State from **In Progress** to **Complete**.
- Save the form.

Activity 5.2 – Export Update Set as XML

The completed Update Set can be exported as an XML file for sharing or submission.

Steps:

- Open **Project Update Set**.
- Under Related Links, click **Export to XML**.
- The Update Set data is downloaded in XML format.
- The downloaded XML file can be shared for transferring the project configuration.



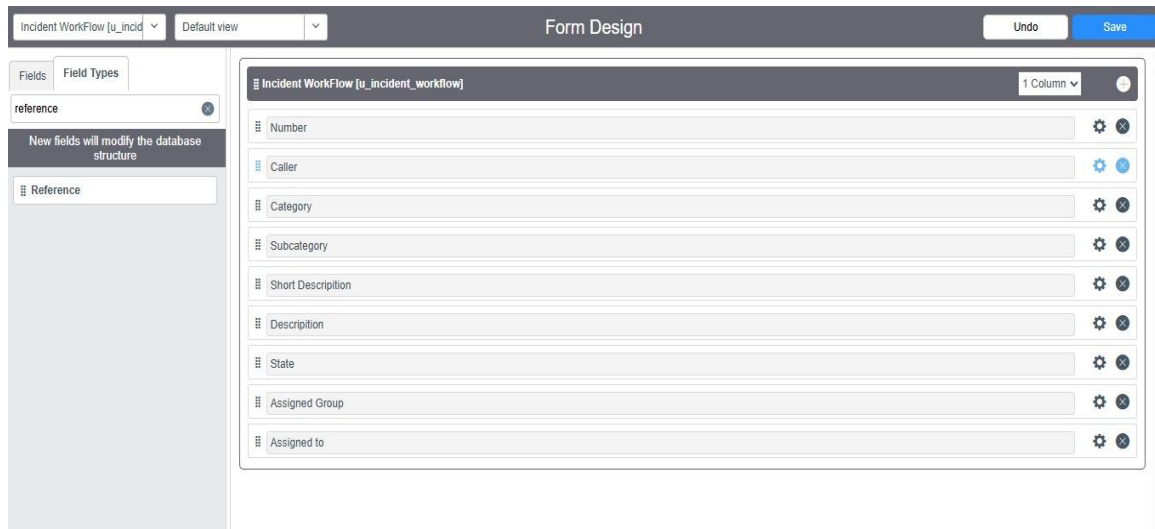
Exploring the ServiceNow Pages

Page 1 – Incident WorkFlow Page

The Incident WorkFlow page is used to create and store IT support ticket records.

The page contains fields such as:

- Number
- Caller
- Category
- Subcategory
- Short Description
- Description
- State
- Assigned Group
- Assigned To



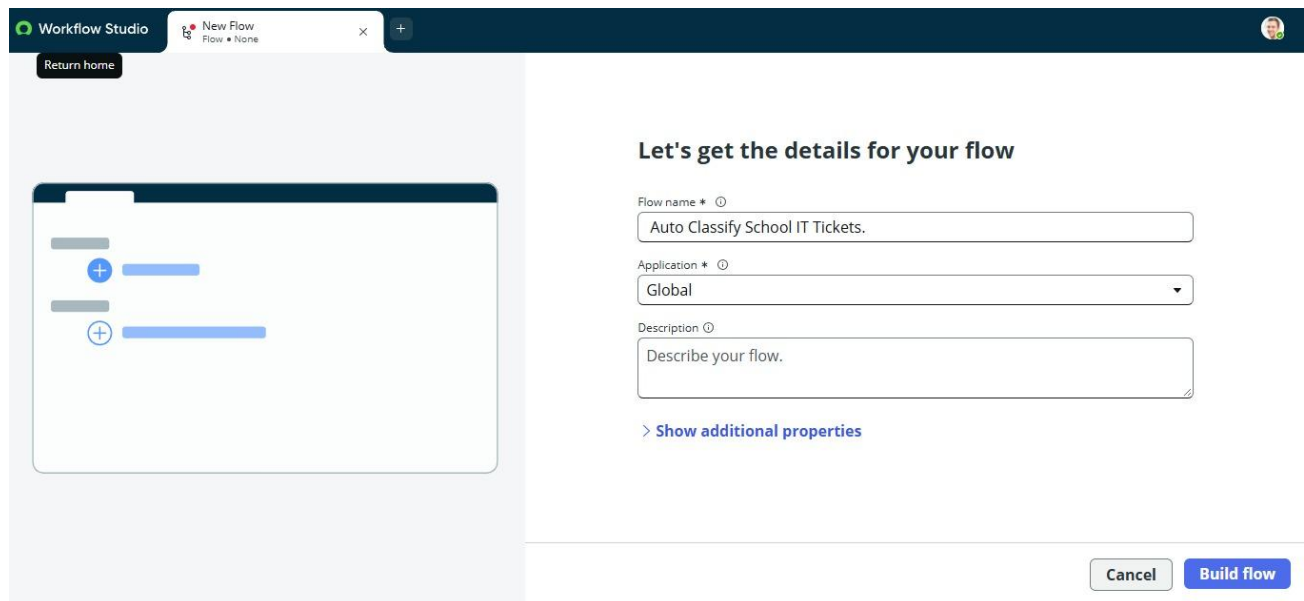
The Form Design interface shows a table with the following fields:

Field Name	Field Type	Field ID
Number	Text	u_incident_workflow_number
Caller	Text	u_incident_workflow_caller
Category	Text	u_incident_workflow_category
Subcategory	Text	u_incident_workflow_subcategory
Short Description	Text	u_incident_workflow_short_description
Description	Text	u_incident_workflow_description
State	Text	u_incident_workflow_state
Assigned Group	Text	u_incident_workflow_assigned_group
Assigned to	Text	u_incident_workflow_assigned_to

Page 2 – Flow Designer Page

The Flow Designer page contains the **Auto Classify School IT Tickets** flow.

The flow contains the record-created trigger and the conditions used to classify the ticket based on the Short Description.



The Workflow Studio interface shows a form for creating a new flow. The form includes the following fields:

- Flow name ***: Auto Classify School IT Tickets.
- Application ***: Global
- Description**: Describe your flow.

Below the form, there is a link: [Show additional properties](#)

At the bottom right, there are two buttons: **Cancel** and **Build flow**.

Page 3 – Classified Incident Page

After the ticket is submitted, the Flow Designer automatically updates the Category and Subcategory according to the issue identified in the Short Description.

For example:

WiFi not working in library

→ **Category: Network**

→ **Subcategory: Wi-Fi**

Projector not turning on

→ **Category: Hardware**

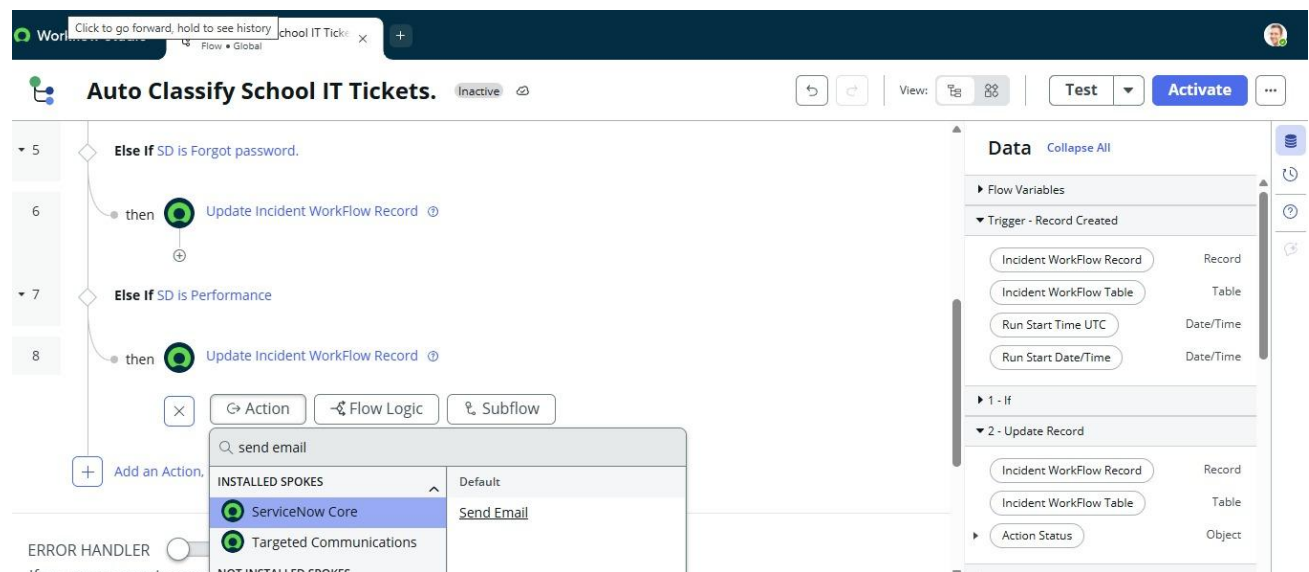
→ **Subcategory: Projector**

Page 4 – Email Notification Page

The Email page displays the notification generated after ticket creation and classification.

The email is addressed to the caller's email address and contains the ticket confirmation

message.



Auto Classify School IT Tickets. Inactive

Click to go forward, hold to see history

Flow • Global

View: [Icons]

Test Activate

5 **Else If** SD is Forgot password.

6 then **Update Incident WorkFlow Record**

7 **Else If** SD is Performance

8 then **Update Incident WorkFlow Record**

+ Add an Action.

ERROR HANDLER

send email

INSTALLED SPOKES

ServiceNow Core

Targeted Communications

Default

Send Email

NOT INSTALLED SPOKES

Data Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record Record

Incident WorkFlow Table Table

Run Start Time UTC Date/Time

Run Start Date/Time Date/Time

1 - If

2 - Update Record

Incident WorkFlow Record Record

Incident WorkFlow Table Table

Action Status Object

Conclusion:

The **Auto Ticket Classification using Flow Designer** project successfully provides an end-to-end automation solution for a school IT helpdesk.

The project automates the classification of IT tickets based on keywords available in the Short Description. The system automatically assigns the appropriate Category and Subcategory without requiring manual classification by IT staff.

The implementation includes the creation of the Incident WorkFlow table, structured ticket fields, dependent Category and Subcategory choices, Flow Designer automation, automatic classification, and email notification.

The solution was tested using different IT support scenarios, including Wi-Fi and projector issues. The test results demonstrated that the appropriate Category and Subcategory were automatically populated and that email notifications were generated for the caller.

The project follows a structured development process consisting of requirement analysis, backend configuration, Flow Designer automation, testing, validation, and Update Set completion. The use of Flow Designer provides a no-code and maintainable approach to implementing the automation.

Overall, the project demonstrates how ServiceNow can be used to automate routine IT helpdesk operations, reduce manual ticket classification effort, improve consistency, and provide a structured approach for managing support requests.

The solution can be extended in the future with additional automation such as automatic assignment, SLA tracking, and predictive intelligence.

